WILLIAM LEE

Senior Backend Engineer - Go, GCP

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PROFESSIONAL SUMMARY

Senior Backend Engineer - Go, GCP with 10+ years delivering secure software across application security, device protection, learning technology, and digital media, using Go 1.13-1.24; combines hands-on architecture, modernization, incident leadership, and mentoring to build scalable systems that improve performance, reliability, security, quality, customer outcomes, and release speed, and long-term engineering productivity.

SKILLS

Languages & Frameworks: Go, SQL, asynchronous programming, dependency injection, background processing
Cloud & Platform: GCP, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, serverless and managed services
Data & Messaging: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Elasticsearch/OpenSearch, Kafka, RabbitMQ, object storage
APIs & Security: REST, GraphQL, gRPC, WebSockets, OpenAPI, OAuth 2.0, OIDC, JWT, mTLS, rate limiting, secrets management
Quality & Observability: unit, integration, contract, end-to-end, load and security testing, OpenTelemetry, Prometheus, Grafana, ELK/OpenSearch
Architecture & Leadership: Microservices, modular monoliths, event-driven architecture, domain-driven design, CQRS where justified, API-first design, ADRs, evolutionary architecture, resilience patterns, distributed transactions, Architecture reviews, technical discovery, estimation, mentoring, code review, incident leadership, Agile delivery, stakeholder communication, roadmap planning

WORK EXPERIENCE

Self-Employed / Independent Consulting | Jun 2026 - Present | Tennessee, United States

Staff Software Engineer

- Architected and delivered **multi-tenant APIs, workflow services and external partner integrations** using **Go 1.22-1.24**, establishing modular boundaries, API contracts, secure defaults, and an incremental roadmap that balanced near-term releases with long-term maintainability.
- Modernized legacy services and user flows into **microservices, event-driven integration, domain-driven design and cloud-native operations**, resolving memory leaks, N+1 database queries, race conditions, stale-cache defects, and timeout failures to reduce **p95 response time by 38%**.
- Built automated delivery on **GCP** with containers, infrastructure as code, environment promotion, secrets management, and rollback controls, shortening **release lead time by 42%** without weakening auditability.
- Designed secure **REST, GraphQL, gRPC, webhook, and event-driven integrations** with OAuth 2.0, OpenID Connect, short-lived credentials, idempotency, rate limiting, and contract-based compatibility controls.
- Introduced unit, integration, contract, end-to-end, performance, and security testing around **Go, GCP**, reducing escaped defects by **31%** and making production releases more predictable across remote teams.
- Optimized relational queries, cache strategy, search indexes, asynchronous workloads, and cloud resource sizing, cutting recurring infrastructure cost by **24%** while preserving availability and growth headroom.
- Led architecture reviews, technical discovery, estimation, code reviews, and incident retrospectives, mentoring engineers through practical design decisions rather than relying on framework-first or rewrite-first solutions.
- Implemented observability with structured logs, metrics, distributed traces, SLO dashboards, and actionable alerts, improving fault isolation and supporting a sustained **99.95% service availability** target.
- Partnered with product, security, data, and operations stakeholders to convert ambiguous requirements into sequenced technical outcomes, documented ADRs, migration plans, and measurable business acceptance criteria.

Veracode | Jul 2021 - May 2026 | Somerville, MA

Staff Software Engineer

- Led architecture and hands-on implementation for **vulnerability ingestion, policy evaluation, scan orchestration and tenant-facing APIs** using **Go 1.16-1.22**, supporting multi-tenant SaaS growth while maintaining tenant isolation, secure coding controls, and backward-compatible customer workflows.
- Refactored tightly coupled modules into **microservices, event-driven integration, domain-driven design and cloud-native operations**, eliminating duplicate processing, connection-pool exhaustion, queue backlogs, and inconsistent retries to improve sustained processing throughput by **34%**.
- Designed versioned APIs and asynchronous contracts for findings, policies, scans, notifications, and external integrations, applying idempotency keys, schema evolution, circuit breakers, and dead-letter recovery.
- Modernized legacy components through staged migrations, compatibility adapters, feature flags, and dual-run validation, avoiding a disruptive rewrite while steadily retiring unsupported libraries and build tooling.
- Strengthened quality gates with static analysis, dependency scanning, unit and integration suites, consumer-driven contract tests, and targeted load tests, reducing escaped production defects by **29%**.
- Improved deployment automation across **GCP** with immutable artifacts, environment policy checks, progressive rollout, and automated rollback, reducing median change lead time by **40%**.
- Diagnosed high-severity production issues involving malformed scan payloads, cross-service timeouts, duplicate events, authorization edge cases, and noisy alerts, then converted findings into durable platform fixes and runbooks.
- Mentored senior and mid-level engineers, facilitated design reviews, and aligned product, security, support, and data teams on technical tradeoffs, service ownership, operational readiness, and measurable delivery outcomes.

Asurion | May 2018 - Jun 2021 | Nashville, TN

Senior Software Engineer

- Developed and evolved **claims intake, device diagnostics, fulfillment, payments and partner integrations** with **Go 1.10-1.16**, designing clear service boundaries and reusable components for customer journeys that had to remain responsive during seasonal and partner-driven traffic spikes.
- Reworked synchronous claim and diagnostic paths into asynchronous, retry-safe workflows with caching and database indexing, reducing average transaction processing time by **27%** and improving peak-load stability.
- Resolved recurring production failures caused by duplicate callbacks, partial partner responses, stale device state, slow queries, and inconsistent error mapping, adding idempotency, reconciliation jobs, and clearer operational telemetry.
- Introduced API versioning, schema validation, OAuth-based authorization, secure secret handling, and contract tests for internal and partner integrations, reducing integration regressions and support escalation volume.
- Expanded automated test coverage across unit, integration, browser or device, and performance scenarios, using CI quality gates and representative test data to reduce customer-reported software defects by **22%**.
- Supported legacy framework and runtime upgrades through dependency audits, compatibility layers, canary releases, and rollback plans, enabling modernization without interrupting claims, diagnostics, or fulfillment operations.
- Collaborated with product, UX, operations, and partner teams in Agile delivery, reviewed designs and pull requests, coached engineers, and translated customer-impacting incidents into prioritized reliability improvements.

Rustici Software | Feb 2016 - Mar 2018 | Nashville, TN

Software Engineer

- Built and maintained **SCORM, xAPI and cmi5 content services, launch workflows and learning-record integrations** using **Go 1.5-1.10**, translating SCORM, xAPI, cmi5, and LMS interoperability requirements into maintainable application modules and testable integration contracts.
- Diagnosed content-launch failures involving malformed manifests, browser security restrictions, session expiration, inconsistent completion state, and vendor-specific payloads, adding validation, compatibility handling, and clearer diagnostics.
- Improved reporting and content-processing performance through query tuning, batch pagination, background workers, cache controls, and selective denormalization while preserving standards compliance and customer data integrity.
- Modernized legacy modules incrementally with automated regression tests, dependency updates, API adapters, and documented migration paths, avoiding broad rewrites that would have increased customer integration risk.
- Implemented REST integrations, webhooks, message-based processing, authentication controls, and structured logging for customer-hosted and SaaS deployments, improving supportability across varied enterprise environments.
- Worked closely with product specialists, support engineers, and customer developers to reproduce difficult interoperability defects, review code, document behavior, and deliver fixes with clear upgrade guidance.

Mediacube | Jun 2015 - Jan 2016 | Knoxville, TN

Junior Software Engineer

- Implemented features for **media ingestion, campaign tracking, billing support and reporting APIs** using **Go 1.2-1.3**, converting product requirements into maintainable application code, SQL changes, background jobs, and user-facing workflows under senior engineering guidance.
- Fixed defects involving duplicate campaign records, failed media uploads, incorrect time-zone calculations, inconsistent totals, and slow reporting queries, adding validation, indexes, transaction boundaries, and actionable logging.
- Created and consumed REST endpoints, integrated third-party media and reporting services, and added retry and error-handling behavior for unreliable external responses and scheduled processing jobs.
- Expanded unit and integration checks, participated in code reviews, documented deployment steps, and supported Linux-based production releases while learning disciplined debugging and change-management practices.
- Collaborated with designers, account teams, and experienced engineers to break work into testable increments, clarify edge cases, and deliver digital-media features without disrupting existing customer reporting.

SELECTED PROJECTS

Multi-Tenant Workflow and Integration Platform

Designed a reference implementation using **Go, GCP** with tenant-aware authorization, versioned REST/gRPC contracts, event-driven workflows, idempotency, PostgreSQL, Redis, Kafka, containerized delivery, and infrastructure as code on **GCP**.

Production Reliability and Developer Portal

Built service templates, deployment automation, API documentation, SLO dashboards, distributed tracing, runbooks, and self-service operational tooling that standardized delivery and reduced the effort required to launch and support new services.

EDUCATION

University of Tennessee, Knoxville | Sep 2011 - May 2013

Bachelor of Science, Computer Science